

Evolved Action-Outcome Comparison in Minimal Recurrent Agents

An exploratory computational methods note | Experiment 06 | 8 October 2026

Scope: sensorimotor computation; not a test or demonstration of consciousness.

Abstract

We tested whether survival-oriented selection creates action-outcome comparison in artificial neural agents initialized without recurrent connections. Across 24 independent evolution seeds and 300 generations per condition, evolved networks achieved a mean expected payoff of 0.1705 (95% bootstrap CI 0.1487–0.1881), against an analytic maximum of 0.2100. A random-selection baseline scored 0.0833 on average. The task was engineered to make earlier motor commands informative about later ambiguous events; accordingly the result demonstrates evolved sensorimotor processing within an artificial task, not subjective experience.

Research design

A 4-unit neural controller saw an action command at time 0, then a binary sensory event at time 1. The event could originate from the action (92% directional correspondence) or an external source (random direction), with equal prior probability. Correctly treating self-caused events as safe and external events as dangerous improved survival payoff. Recurrent links capable of carrying the earlier command were absent initially and could arise by mutation; no explicit self/other label was available to the agent.

Methods and analyses

24 independent seeds; population 192; 300 generations; truncation selection, genotype inheritance, structural and continuous mutations. Same-budget baselines: no recurrence, random fitness selection, no genotype inheritance. Exact expected survival payoff was the primary metric. Secondary: hold-out Monte Carlo trials, memory ablation, cue scrambling, contingency change, and post-hoc causal-source AUC. Analysis plan frozen locally, NOT externally preregistered.



Figure 1. Each dot is a separate evolutionary run. Black markers show condition means and 95% bootstrap intervals; dotted line indicates theoretical maximum.

Findings and qualifications

Results

Condition	Mean payoff	95% bootstrap CI
Evolved recurrence	0.1705	0.1487 to 0.1881
No recurrence	0.0000	0.0000 to 0.0000
Random selection	0.0833	0.0691 to 0.0971
No inheritance	0.0000	0.0000 to 0.0000

Post-hoc AUC for hidden event source: 0.631 (chance 0.50). On a reduced-reliability environment ($q=0.80$), expected payoff was 0.1218; when no information was present ($q=0.50$), 0.0000; with inverted contingency ($q=0.08$), -0.1705.

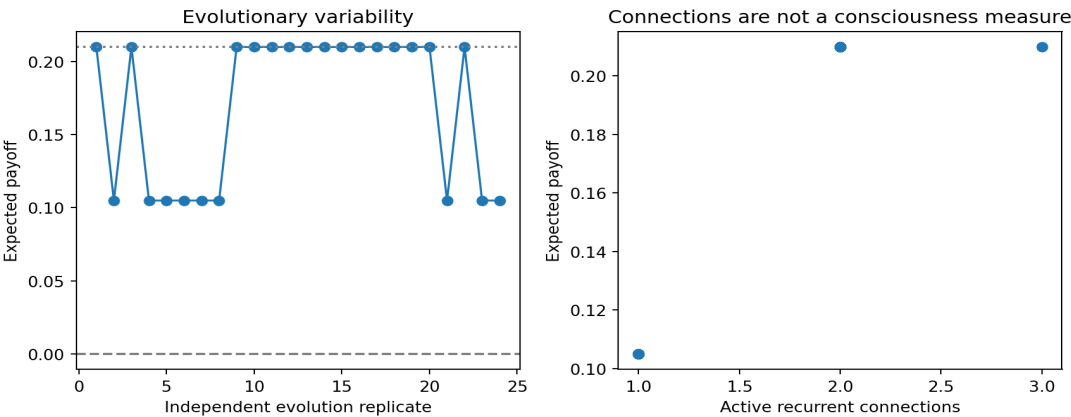


Figure 2. Runs vary substantially. Connections are structural prerequisites here, not measures of experience.

What can be concluded

A recurrent action-history pathway can evolve from initially absent connections and improve survival in an engineered two-stage motor/sensory task. The behavior is contingent on the environmental model and disappears when the historical information is made inaccessible. This is a minimal computational illustration of action-outcome comparison.

What cannot be concluded

No agent was shown to have first-person experience, awareness, selfhood, a biological brain, or an independently developed theory of its own body. The environment, actions, payoffs, possible neurons, and mutation process were designed by the investigator. The principle overlaps substantially with known corollary-discharge and robotics research. Do not present the work as a consciousness discovery.

Publication assessment

Appropriate for an open, transparently labeled computational methods note or educational demonstration once authorship and licensing are settled; not yet established as a novel, peer-review-ready contribution. Recommended next steps: independent replication, new and more ecological environments, alternative agent architectures, sensitivity analysis, comparison against published baselines, and public preregistration before any new confirmatory study.

Experiment 06 — Publication Audit Addendum

Post-hoc methodological review | Experimental computational model | Not peer-reviewed

What the original simulation demonstrates

Across 24 independent evolution seeds, selected recurrent controllers achieved mean exact expected payoff 0.1705 versus 0.0833 under random-fitness selection (paired mean difference 0.0872; 95% bootstrap interval 0.0611–0.1130). This demonstrates the usefulness of a mutation-enabled action-history link in a deliberately constructed environment. It does not measure consciousness or a genuine self-model.

Important control flaw

The original no-inheritance condition also prevents recurrent-link mutation. It cannot isolate the effect of inheritance. Its zero-score outcome must not be interpreted as an independent proof that inheritance causes self-modeling.

Post-hoc follow-up baseline: fresh genomes with possible links

A new, separately labelled sensitivity check regenerated 192 unrelated agents in each generation and allowed each fresh genotype to have recurrent links. Selection of the best final-generation controller used the same analytic survival criterion. This deliberately simplified control is not a rerun of the original evolutionary process.

Condition	Mean payoff	95% bootstrap interval
Original evolved	0.17051	0.14865–0.18812
NEW de novo no-inheritance	0.02030	0.01766–0.02323
Paired difference	0.15022	0.12943–0.16933

Additional weaknesses

Training optimizes exact payoff using the known environment distribution; the motor command is externally supplied rather than chosen; champions are chosen post hoc by highest true task payoff even for the random-fitness control; the task and possible recurrent architecture are engineered; there is no independent replication or public preregistration.

Release recommendation

Share as an exploratory model only, with all code and the methodological addendum. For academic submission, preregister a new ecological simulation with stochastic selection, agents choosing their actions, clean mutation and inheritance controls, multiple architecture families, and independent external replication.

Research context

Crapse & Sommer (2008), DOI: 10.1038/nrn2457; Bongard et al. (2006), DOI: 10.1126/science.1133687; Tanaka & Imamizu (2025), DOI: 10.1038/s44271-025-00240-7. The general phenomenon predates this simulation.